2026 BROCK AI DAY · EXPLAINABLE AI / DEEP LEARNING

Are language models converging on shared structures?

Mechanistic interpretability, the linear representation hypothesis, and the platonic representation hypothesis

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1 · CONCEPTS ARE VECTORS

A model stores each concept as a direction

WITH CREDIBILITY MARKERS

“Traditional vaccine development is a meticulous process, typically requiring 10–15 years…”

SAME CLAIM, NO GROUNDING

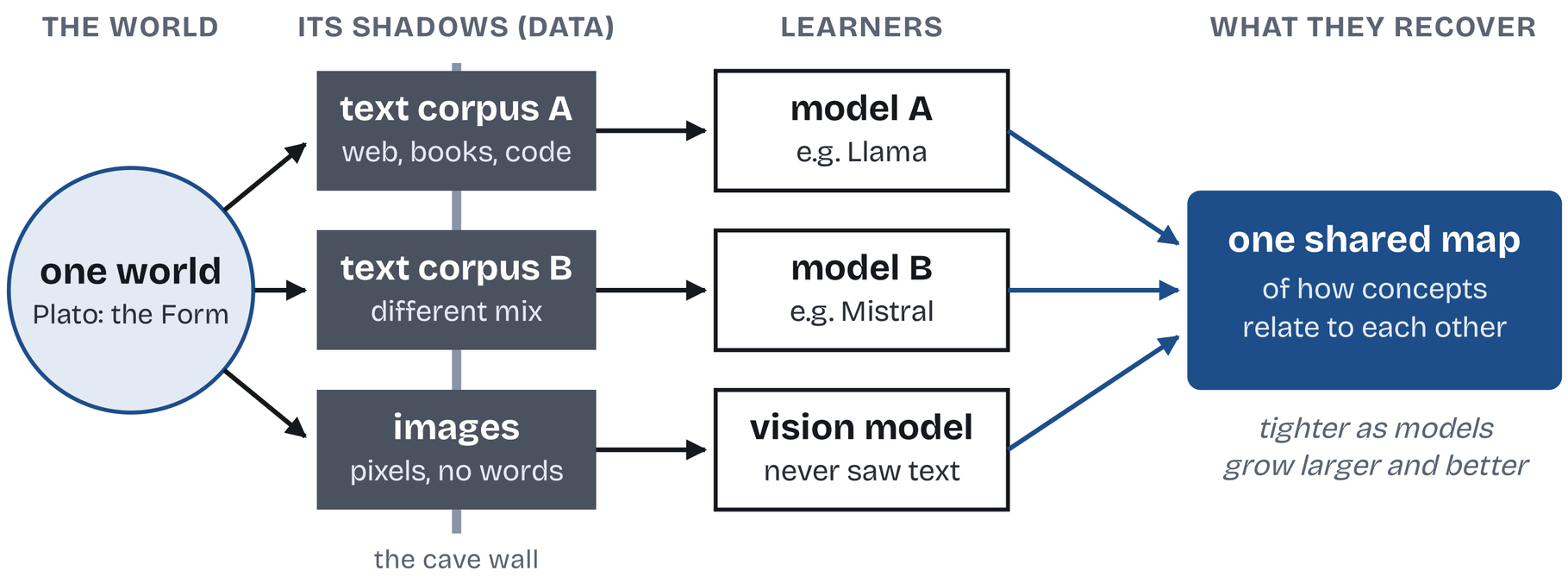
“It’s just common sense that a vaccine should take years to develop…”

mean(with) − mean(without) = the concept vector, at every layer

17 concepts · about 250 pairs each · written by 14 different LLMs

2 · THE HYPOTHESIS · HUH, CHEUNG, WANG & ISOLA, 2024

Different shadows, one world?

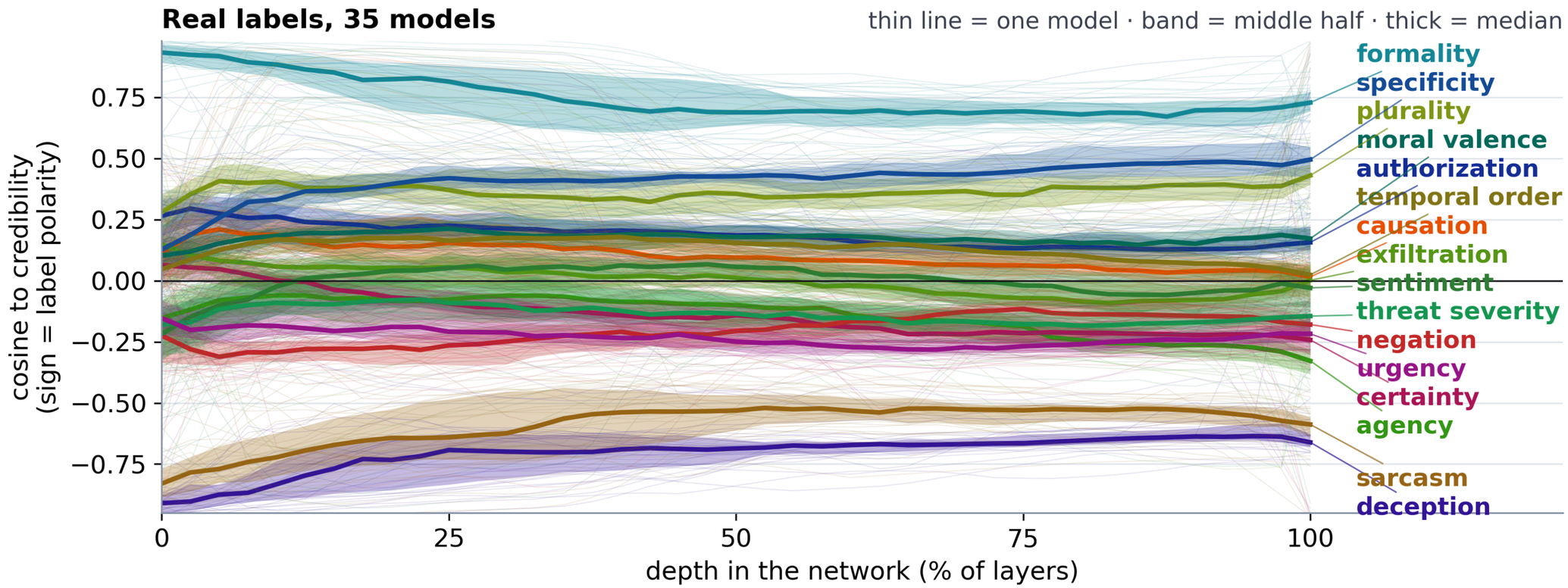


What converges is relations, not coordinates.

Prediction: models that never shared a weight should agree on how concepts relate to each other.

3 · OUR TEST: 35 MODELS, EVERY LAYER

What is credibility aligned with?



Each line is a concept's cosine to credibility across depth; bands show the spread over 35 models. At the final layer formality is closest in 29 of 35. Deception and sarcasm are nearly as strong with the opposite sign (sign = label polarity).

4 · THE TEST THAT CAN FAIL

Can angles alone name the concepts?

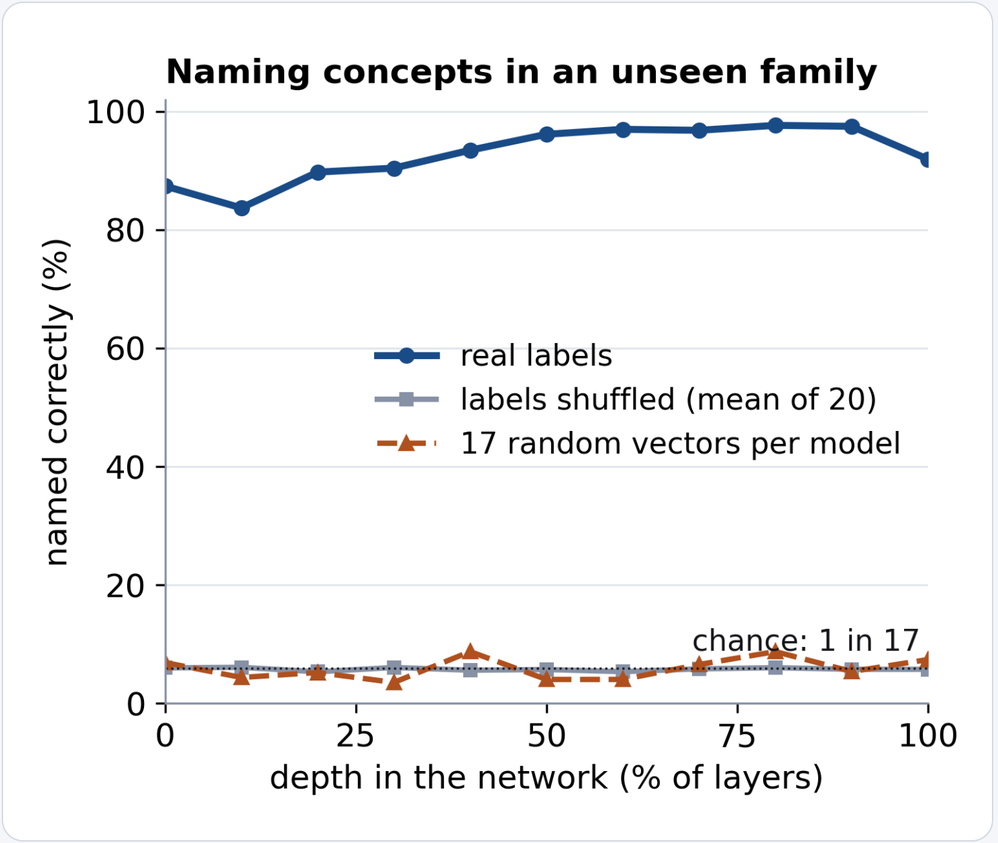
97%

identified correctly at mid-depth

6%

with shuffled labels or random vectors: chance

Hold out a whole model family. Describe each of its concept vectors only by its angles to the others, then match against the average from the other families.



WHAT IT MIGHT MEAN

The consolidation of meaning

Thirty-five models, built by different teams on different text, agree on how their concepts sit relative to one another. Each model's coordinates are its own; the angles between its concepts are not.

Is this the structure of language itself, or the structure of the world it describes?

Poster source, figure code and reading list: github.com/jamesrahenry/ai-day-2026